

# Data Integration Report

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## Abstract

We construct a unified training dataset by merging JARVIS-DFT and Materials Project via a structure-aware classification scheme. Layered/2D materials retain JARVIS OptB88vdW labels, while 3D bulk materials are preferentially assigned Materials Project PBE values. After deduplication, the final dataset comprises 165,533 entries, ensuring each normalised formula appears exactly once.

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## 1 Data Sources Used

We combine two complementary DFT databases: JARVIS-DFT, computed with the van-der-Waals-corrected OptB88vdW functional and therefore well-suited to layered and 2D materials, and Materials Project, computed at the PBE level with substantially broader coverage of 3D bulk and metallic solids. Rather than naively concatenating both databases, we apply a structure-aware “smart merge”: JARVIS materials are first classified as either layered/2D/van-der-Waals solids (Group A) or 3D bulk/metallic/closely-packed solids (Group B), using exfoliation energy, density, and crystal system as discriminating features (JARVIS is the only one of the two sources reporting exfoliation energy directly). Group A materials retain their OptB88vdW-level JARVIS labels (21,537 entries). Group B materials are preferentially re-labelled with PBE-level Materials Project values when an MP entry for the same normalised formula exists (134,100 entries), falling back to the JARVIS entry otherwise (9,895 entries). After deduplication to one row per `formula_norm` (lowest `energy_above_hull_eV` retained), the final dataset totals 165,533 entries with formation energy, band gap, energy above hull, crystal system, space group, density and composition fields populated for every row. Source-level statistics are summarised in Table 1.

Table 1: Summary of data sources included in the training dataset.

| Source            | Raw entries | Final entries | Properties                                               | Functional |
|-------------------|-------------|---------------|----------------------------------------------------------|------------|
| Materials Project | 210,579     | 134,100       | $E_f$ , $E_g$ , $E_{hull}$ , density, crystal system     | PBE (GGA)  |
| JARVIS-DFT        | 77,096      | 31,432        | $E_f$ , $E_g$ , $E_{hull}$ , exfoliation energy, density | OptB88vdW  |

### 1.1 Materials Project

Entries were retrieved via the MP API with full provenance tracking (`mp_properties_with_provenance.csv`), yielding 210,579 raw records. For each material we collected the normalised formula (`formula_norm`), composition, band gap (`bandgap_eV`), formation energy per atom (`formation_energy_eV_per_atom`), energy above hull (`energy_above_hull_eV`), crystal system, space group number, and density. All MP values are computed at the PBE (GGA) level. As is standard for MP, band gaps reported here are GGA values and are therefore systematically underestimated relative to experiment; no HSE06-corrected gaps were used in this dataset. After deduplication and the merge logic described in Section 2, 134,100 of these records (Group B, those sharing a `formula_norm` with a JARVIS entry, plus MP-exclusive formulas) were retained in the final dataset.

### 1.2 JARVIS-DFT

We used the JARVIS complete database export (`JARVIS_COMPLETE_DATABASE.csv`), comprising 77,096 raw records and providing formation energy, band gap, energy above hull, crystal system, space group, density, and - critically for our classification step - exfoliation energy, a quantity not available from Materials Project. JARVIS values are computed with the OptB88vdW van-der-Waals-corrected functional, which makes it the more physically appropriate source for layered and 2D materials, where dispersion interactions between layers are not well captured by standard PBE. After classification and deduplication, 31,432 JARVIS-sourced records were retained in the final dataset: 21,537 as Group A (layered/2D, JARVIS labels kept) and 9,895 as Group B fallback (3D-classified materials with no Materials Project match for their `formula_norm`).

### 1.3 Additional Sources

No additional databases beyond Materials Project and JARVIS-DFT were used.

## 2 Harmonisation Strategy

### 2.1 Structure Matching

Matching between the two databases is performed on `formula_norm` (normalised reduced formula). For every JARVIS entry assigned to Group B (see below), we search Materials Project for an entry sharing the same `formula_norm`; if found, the MP row replaces the JARVIS row. MP entries whose `formula_norm` does not appear among JARVIS entries at all are appended directly to the dataset as Group B. The breakdown by stage is summarised in Table 2.

Table 2: Entry counts at each stage of the merge pipeline.

| Group                              | Source                             | Entries        |
|------------------------------------|------------------------------------|----------------|
| A                                  | JARVIS-DFT (layered/2D)            | 21,537         |
| B                                  | JARVIS-DFT (no MP match, fallback) | 9,895          |
| B                                  | Materials Project                  | 134,100        |
| <b>Total (final, deduplicated)</b> |                                    | <b>165,533</b> |

Out of 77,096 raw JARVIS records and 210,579 raw MP records, deduplication (Section 2.3) and the classification/matching logic above reduce the union to 165,533 unique materials, one row per `formula_norm`.

## 2.2 Classification Logic (Group A vs. Group B)

Because only JARVIS reports exfoliation energy and density, classification is performed on JARVIS entries and is used to decide, for each formula, which functional/source is physically more appropriate:

Table 3: Structural classification rules applied to JARVIS entries.

| Priority | Condition                                                               | Group | Interpretation                                                                                  |
|----------|-------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------|
| 1        | Exfoliation energy < 150 meV/atom and Density $\leq 4.0 \text{ g/cm}^3$ | A     | Weak vdW binding $\Rightarrow$ genuinely layered/2D material (easily exfoliable).               |
| 2        | Exfoliation energy $\geq 150 \text{ meV/atom}$                          | B     | Strongly bound layers $\Rightarrow$ effectively 3D-like stacking (Non-exfoliable bulk material) |
| 3        | Crystal system trigonal or hexagonal                                    | A     | Structural tiebreaker favouring layered symmetry.                                               |
| 4        | Default                                                                 | B     | Treated as 3D bulk/metallic.                                                                    |

The threshold values of 150 meV/atom for exfoliation energy and  $4.0 \text{ g/cm}^3$  for density were selected based on the expert recommendation of a team member holding a PhD in Chemistry, who validated these boundaries against established computational literature and empirical distributions of van-der-Waals binding energies and bulk densities for layered versus close-packed solids.

Group A materials retain their native JARVIS OptB88vdW labels. Group B materials are preferentially re-labelled with Materials Project PBE values (Section 2.1); where no MP match exists, the JARVIS row is kept as a fallback so that no material is dropped from the dataset purely due to missing MP coverage.

For the 134,100 Group B rows sourced from Materials Project, density and crystal system are taken directly from the MP API, as these properties are natively provided in the MP dataset. However, exfoliation energy is not available in Materials Project - it is a JARVIS-specific property. Therefore, exfoliation energy is used exclusively as a classification criterion for JARVIS-DFT entries (Group A vs. Group B) and is not populated or used as a feature for MP-origin rows in the final training set.

## 2.3 Handling Duplicates

After merging, the dataset is deduplicated to exactly one row per **formula\_norm**. Where multiple polymorphs/entries exist for the same normalised formula, the entry with the lowest **energy\_above\_hull\_eV** is retained, ensuring the dataset represents the most thermodynamically stable known structure for each composition.

## 2.4 Label Assignment

The primary label source is therefore composition- and structure-dependent rather than fixed: JARVIS (OptB88vdW) is primary for Group A (layered/2D/vdW) materials, and Materials Project (PBE) is primary for Group B (3D bulk/metallic) materials, with JARVIS used as a fallback for Group B when no MP entry is available. This choice reflects the physical regimes in which each functional is most reliable - OptB88vdW’s dispersion correction benefits weakly-bound layered systems, while PBE is the long-established, widely benchmarked standard for dense 3D solids - rather than treating one database as uniformly authoritative.

## 2.5 Methodological Bias Correction

No linear bias correction between MP and JARVIS formation energies/band gaps was applied, and none was needed: the JARVIS and Materials Project subsets of the final dataset are disjoint by construction (Group A and Group B are assigned per formula, with each formula contributing exactly one row from exactly one source), so there is no set of materials with both an MP and a JARVIS label to compare or reconcile. Each material in the final dataset therefore carries labels from a single functional, consistent with the structural regime it was assigned to (Section 2.4).

## 3 Data Card

Overlap in the final dataset: none - all conflicting **formula\_norm** entries present in both raw databases were resolved by the Group A/B priority rule (Section 2.2), ensuring each formula appears exactly once in

Table 4: Dataset statistics (training split).

| Property                             | Value                                                       |
|--------------------------------------|-------------------------------------------------------------|
| Total entries                        | 165,533                                                     |
| from JARVIS-DFT (Group A + fallback) | 31,432                                                      |
| from Materials Project (Group B)     | 134,100                                                     |
| Total entries (train / val)          | 132,426 / 33,107 (80% / 20%)                                |
| Test set                             | held out, evaluated by organisers                           |
| Unique chemical systems              | 165,533                                                     |
| Overlap in final dataset             | None (resolved by Group A/B priority rule; see Section 2.4) |
| Splitting strategy                   | Random 80/20 train/val split                                |

the merged set. This does not imply the raw JARVIS and MP databases themselves are disjoint; rather, every overlap between them was deliberately resolved during merging.

Table 5: Preprocessing applied per source in the final dataset.

| Source                          | Preprocessing applied                                                                                                                                                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JARVIS-DFT (Group A + fallback) | Classification by exfoliation energy/density/crystal symmetry; deduplication by lowest <code>energy_above_hull_eV</code> ; OptB88vdW labels retained.                                                                         |
| Materials Project (Group B)     | Normalised-formula matching against JARVIS; deduplication by lowest <code>energy_above_hull_eV</code> ; PBE labels used; density and crystal system extracted natively from MP; exfoliation energy marked N/A for these rows. |

Histograms of formation energy and band gap for the training set, generated from `bridge_dataset_train.csv`, can be inserted here as a figure.

## 4 Limitations and Failure Cases

- **Mixed-functional dataset.** Because Group A materials carry OptB88vdW labels and Group B materials carry PBE labels (or OptB88vdW as fallback), the dataset is not computed at a single, uniform level of theory. A model trained on it implicitly learns a blend of two functionals, which may introduce systematic error at the Group A/B decision boundary. Since the two source subsets are disjoint (no shared materials), this discrepancy cannot be empirically quantified or corrected from the data itself.
- **Heuristic classification boundary.** The 150 meV/atom exfoliation-energy threshold and the density/crystal-system tiebreakers (Table 3) are reasonable physical heuristics but are not a rigorous classifier; materials near the threshold may be assigned to the “wrong” group, and borderline cases (e.g. structures with null exfoliation energy and density just above/below 4.0 g/cm<sup>3</sup>) are especially sensitive to this choice.
- **Exfoliation energy unavailable for MP-origin rows.** Density and crystal system are natively reported by both sources and are populated for every row. Exfoliation energy, however, is a JARVIS-specific quantity with no Materials Project equivalent; it is used only as a classification criterion for JARVIS entries (Section 2.2) and is marked N/A for the 134,100 Group B rows sourced from Materials Project. Any downstream use of exfoliation energy as a model feature is therefore restricted to the JARVIS-origin subset of the dataset.
- **Band gap reliability.** Both PBE and OptB88vdW are GGA-level functionals and are known to systematically underestimate band gaps relative to experiment or hybrid-functional (HSE) calculations; semiconductors and insulators in this dataset should be expected to have under-predicted gaps, and some narrow-gap materials may be mislabelled as metallic.
- **Coverage imbalance.** The final dataset is dominated by Materials Project (134,100 of 165,533 entries, 81%); JARVIS contributes only 31,432 entries (19%), all of which are either explicitly

layered/2D materials (Group A) or 3D materials absent from MP (Group B fallback). Property distributions for layered/2D systems are therefore drawn from a comparatively small, JARVIS-only slice of the data, which may limit model performance specifically on van der Waals materials relative to bulk 3D materials.

- **No materials explicitly excluded.** All 77,096 raw JARVIS records and all 210,579 raw MP records were used as input to the classification, matching, and deduplication pipeline; no records were manually removed prior to merging. The reduction to 165,533 final rows results entirely from deduplication to one row per `formula_norm` and from the Group A/B merge logic, not from any additional filtering criteria.

## References

1. Jain et al., *APL Mater.* **1**, 011002 (2013) — Materials Project.
2. Choudhary et al., *npj Comput. Mater.* **6**, 173 (2020) — JARVIS-DFT.